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Inventor(s)	Murota; Masaya et al.

Switch device

Abstract

A switch device includes, a movable terminal including a snap action mechanism that bends a spring when a user moves an operating portion, a first fixed terminal that contacts the movable terminal when the spring is not bent, a second fixed terminal that contacts the movable terminal when the operating portion is moved by a specified amount to bend the spring, a case including an opening, and a cover that closes the opening and is separate from the first fixed terminal. When the opening is closed by the cover, the first fixed terminal is positioned in the case by the cover.

Inventors: Murota; Masaya (Aichi, JP), Fujimoto; Takeshi (Aichi, JP)

Applicant: KABUSHIKI KAISHA TOKAI RIKA DENKI SEISAKUSHO (Aichi, JP)

Family ID: 88857148

Assignee: KABUSHIKI KAISHA TOKAI RIKA DENKI SEISAKUSHO (Niwa-gun, JP)

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Primary Examiner: Jimenez; Anthony R

Background/Summary

BACKGROUND

1. Field

(1) The following description relates to a switch device that switches contacts with a snap action mechanism.

2. Description of Related Art

(2) Japanese Laid-Open Patent Publication No. 2010-73662 discloses a known switch device including a snap action mechanism that is actuated when pressed by an operation member. The switch device of Japanese Laid-Open Patent Publication No. 2010-73662 includes a housing, the operation member, a fixed contact, a movable contact, and the snap action mechanism. The snap action mechanism drives the movable contact when the operation member is pressed to a predetermined position.

SUMMARY

(3) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in

determining the scope of the claimed subject matter.

(4) In one general aspect, a switch device includes a movable terminal including a snap action mechanism that bends a spring when a user moves an operating portion, a first fixed terminal that contacts the movable terminal when the spring is not bent, a second fixed terminal that contacts the movable terminal when the operating portion is moved by a specified amount to bend the spring, a case including an opening, and a cover that closes the opening and is separate from the first fixed terminal. When the opening is closed by the cover, the first fixed terminal is positioned in the case by the cover.

(5) Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a switch unit according to one embodiment.

(2) FIG. 2 is a perspective view of a switch device.

(3) FIG. 3 is an exploded perspective view of the switch device.

(4) FIG. 4A is a diagram showing an operation of the switch device.

(5) FIG. 4B is a diagram showing an operation of the switch device.

(6) FIG. 5 is a perspective view showing how a first fixed terminal is coupled to a case.

(7) FIG. 6 is a side view of the switch device.

(8) FIG. 7 is a cross-sectional view taken along line VII-VII in FIG. 6.

(9) FIG. 8 is a cross-sectional view taken along line VIII-VIII in FIG. 2.

(10) FIG. 9 is a cross-sectional view taken along line IX-IX in FIG. 6.

(11) FIG. 10 is a partially enlarged perspective view of the switch device.

(12) FIG. 11 is a perspective view showing the inside of the case where a fixed contact is arranged.

(13) FIG. 12 is a perspective view showing the inside of the case where the fixed contact is arranged.

(14) FIG. 13 is a diagram showing the electric configuration of a switch unit.

(15) FIG. 14 is a table for determining the state of contacts.

(16) FIG. 15 is a diagram illustrating the switch device reduced in size.

(17) Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

(18) This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

(19) Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

(20) In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

(21) One embodiment of the present disclosure will now be described.

(22) Switch Unit 1

(23) As shown in FIG. 1, a switch unit 1 is, for example, an electric parking brake switch 2 that is

manually moved to electrically activate and deactivate the parking brake of a vehicle. The switch unit **1** is located, for example, inside the passenger compartment (near driver seat) of the vehicle. (24) The switch unit **1** includes a frame **3** and an operating portion **4** movably attached to the frame **3**. The operating portion **4** attached to the frame **3** is rotatable about a shaft **5** arranged in, for example, the frame **3** or the operating portion **4**. The operating portion **4** can be rotated in two directions (first operation direction **R1**, second operation direction **R2**) about the axis **La** of the shaft **5**. The operating portion **4** is rotated, for example, in the first operation direction **R1** or the second operation direction **R2** from a neutral position. The operating portion **4** is, for example, of a momentary type in which the operating portion **4** automatically returns to the neutral position when a user releases the operating portion **4** after rotating it from the neutral position.

(25) To activate the parking brake, for example, the user moves the operating portion **4** in the first operation direction **R1** from the neutral position while depressing the braking pedal. Movement of the operating portion **4** in the first operation direction **R1**, for example, raises the operating portion **4**. In this case, the parking brake is activated when the operating portion **4** is moved to a first operation position (electric parking brake activation position) at the limit end in the first operation direction **R1**.

(26) To deactivate the parking brake, for example, the user moves the operating portion **4** in the second operation direction **R2** from the neutral position while depressing the braking pedal. Movement of the operating portion **4** in the second operation direction **R2**, for example, lowers the operating portion **4**. In this case, the parking brake is deactivated when the operating portion **4** is moved to a second operation position (electric parking brake deactivation position) at the limit end in the second operation direction **R2**. The parking brake may be automatically deactivated when, for example, a shifter of the vehicle is set to the drive position and the vehicle starts to travel.

(27) Switch Device **8**

(28) As shown in FIG. 2, the switch unit **1** includes switch devices **8** that detect movement of the operating portion **4**. The switch devices **8** of the present example include a first switch device **8a** that detects movement of the operating portion **4** in the first operation direction **R1** and a second switch device **8b** that detects movement of the operating portion **4** in the second operation direction **R2**. The two switch devices **8** are, for example, arranged next to each other. The switch devices **8** basically have the same structure. The structure of the switch devices **8** will now be described.

(29) The switch device **8** includes a housing **9** that accommodates device components. The housing **9** is substantially a rectangular parallelepiped. The housing **9** includes a case **11**, having an opening **10**, and a cover **12** that closes the opening **10**. The case **11** is, for example, an insulator that insulates electric components of the switch device **8**. The case **11** is box-shaped and has a closed end and an opposite end (upper end in present example) that is closer to the cover **12** and includes the opening **10**. In one example, the case **11**, which is box-shaped, includes a bottom surface that is closed and an upper surface that is open by the opening **10**. The case **11** includes two opposing side walls **13**. The cover **12** is fixed to the case **11** through, for example, welding. The welding may be laser welding.

(30) As shown in FIG. 3, a leg **14** extends from each of the four corners of the rear surface of the cover **12**. The four corners in the open end of the case **11** each include an engagement portion **15** paired with a corresponding one of the legs **14** of the cover **12**. The cover **12** is coupled to the case **11** so that the legs **14** are engaged with the corresponding engagement portions **15** of the case **11**.

(31) Each switch device **8** includes a movable terminal **18**, a first fixed terminal **19**, and a second fixed terminal **20**. The movable terminal **18** is moved when the user moves the operating portion **4**. Further, the movable terminal **18** selectively contacts the first fixed terminal **19** and the second fixed terminal **20**. The movable terminal **18** includes a movable terminal body **18a** and movable terminal connection pins **18b** mounted on a board **21**. Two movable terminal connection pins **18b** are respectively arranged, for example, at the two sides of the switch device **8** in the width direction (device width direction: Y-axis direction in FIG. 3).

(32) A linking portion **18c**, used when forming the movable terminal **18** during a pressing process, is arranged adjacent to each movable terminal connection pin **18b**. The linking portion **18c** does not have the functionality of a terminal and may be omitted.

(33) The movable terminal connection pins **18b** extend out of the case **11** through holes **11a** formed in the side walls **13** of the case **11**. The linking portions **18c** extend out of the case **11** through holes **11b** formed in the side walls **13** of the case **11**.

(34) The movable terminal **18** has a snap action mechanism **23** that bends a spring **22** when the user moves the operating portion **4**. The proximal end of the spring **22** is fixed to two engaging portions **24**, **25** arranged on the movable terminal body **18a**. The distal end of the spring **22** includes a movable contact **26** of the movable terminal **18**. The movable contact **26** is, for example, a rivet. The rivet of the present example undergoes, for example, a swaging process for attachment to the spring **22**. The rivet has the minimum diameter that allows for swaging.

(35) The spring **22** includes a tensile portion **27** that applies tensile force to the proximal end of the spring **22** and a compression portion **28** that applies reaction force in a direction opposite to the direction of tensile force. The proximal end of the tensile portion **27** is engaged with the engaging portion **24**, which is located toward one end of the case **11**. The proximal end of the compression portion **28** is engaged with the engaging portion **25**, which is located at the middle part of the case **11**.

(36) A pressing portion **31** moved in cooperation with the operating portion **4** is arranged on the proximal end of the tensile portion **27** opposing the operating portion **4**. The pressing portion **31** includes a head **31a** that is exposed to the outside of the housing **9** through a hole **32** extending through the cover **12**. The pressing portion **31** is moved back and forth in the height direction (device height direction: Z-axis direction in FIG. **3**) of the switch device **8**. The spring **22** bends when the pressed amount of the pressing portion **31** reaches a specified amount.

(37) The first fixed terminal **19** includes a first fixed terminal body **19a**, which contacts the movable terminal **18**, and first fixed terminal connection pins **19b**, which are arranged at the ends of the first fixed terminal body **19a**. The first fixed terminal **19** is located closer to the cover **12** than the movable terminal **18**. The first fixed terminal body **19a** is, for example, C-shaped. The first fixed terminal body **19a** is located at a position opposed to the movable terminal **18**.

(38) The first fixed terminal connection pins **19b** are arranged at, for example, the two ends of the first fixed terminal body **19a**. The first fixed terminal connection pins **19b** are pin-shaped and have a narrower width than the first fixed terminal body **19a**. The first fixed terminal connection pins **19b** extend out of the case **11** through groove portions **33** formed in the side walls **13** of the case **11**. The first fixed terminal connection pins **19b** are mounted on the board **21**.

(39) The second fixed terminal **20** includes a second fixed terminal body **20a** that contacts the movable terminal **18** and second fixed terminal connection pins **20b** arranged at the ends of the second fixed terminal body **20a**. The second fixed terminal **20** is located closer to the bottom of the case **11** than the movable terminal **18**. The second fixed terminal body **20a** is located at a position opposed to the movable terminal **18**. The second fixed terminal connection pins **20b** are arranged at, for example, two opposite ends of the second fixed terminal body **20a**. The second fixed terminal connection pins **20b** are mounted on the board **21**.

(40) A linking portion **20c**, used when forming the second fixed terminal **20** during a pressing process, is arranged adjacent to each second fixed terminal connection pin **20b**. The linking portion **20c** does not have the functionality of a terminal and may be omitted.

(41) The second fixed terminal connection pins **20b** extend out of the case **11** through holes **11d** formed in the side walls **13** of the case **11**. The linking portions **20c** extend out of the case **11** through holes **11e** formed in the side walls **13** of the case **11**.

(42) The movable terminal **18** (excluding spring **22**) and the second fixed terminal **20** may be fixed to the case **11** from the lower side. Alternatively, the movable terminal **18** (excluding spring **22**) and the second fixed terminal **20** may be insert-molded to the case **11**.

(43) Action for Switching Contacts in Switch Device **8**

(44) As shown in FIG. 4A, when the operating portion **4** is not moved and located at the neutral position, the spring **22** of the snap action mechanism **23** is not bent. In this case, the resultant force of the tensile force of the tensile portion **27** and the reaction force of the compression portion **28** in the spring **22** acts upward (toward first fixed terminal **19**) on the movable contact **26**. Thus, the movable contact **26** of the movable terminal **18** contacts the first fixed terminal **19**.

(45) As shown in FIG. 4B, when the operating portion **4** is rotated in the first operation direction **R1** (second operation direction **R2**) thereby pressing the pressing portion **31** so that the operating portion **4** is moved for the specified moved amount, the resultant force of the tensile force and the reaction force acts downward (toward second fixed terminal **20**) on the movable contact **26**. Thus, the movable contact **26** moves away from the first fixed terminal **19** and contacts the second fixed terminal **20**. That is, the connection of the movable terminal **18** switches from the first fixed terminal **19** to the second fixed terminal **20**.

(46) When the operating portion **4** is released, the spring **22** bends back to its original state. Thus, the movable contact **26** returns to the state contacting the first fixed terminal **19**. In this manner, the movable contact **26** selectively contacts the first fixed terminal **19** and the second fixed terminal **20** in accordance with the moved amount of the operating portion **4**.

(47) Structure of First Fixed Terminal **19**

(48) As shown in FIG. 3, the first fixed terminal **19** of the present example is separate from, for example, the cover **12**. The first fixed terminal **19** is positioned in the case **11** by the cover **12** when, for example, the opening **10** is closed by the cover **12**. Specifically, the first fixed terminal **19** is pressed from above by the cover **12** and positioned when coupled to the case **11**.

(49) The first fixed terminal connection pins **19b** are bent in advance in the direction parallel to a mounting surface **34** of the board **21** before the first fixed terminal **19** is coupled to the case **11**. In this manner, the first fixed terminal connection pins **19b** are bent in advance. The first fixed terminal connection pins **19b** of the present example extend in the horizontal direction (X-Y plane direction in FIG. 3) on the mounting surface **34** of the board **21**. Thus, the first fixed terminal connection pins **19b** are bent in advance when the first fixed terminal **19** is coupled and fixed to the case **11**.

(50) Structure of Groove Portions **33**

(51) As shown in FIGS. 5 and 6, when coupling the first fixed terminal **19** to the case **11**, the groove portions **33** allow for passage of the first fixed terminal connection pins **19b**. The groove portions **33** of the present example are formed in the two side walls **13**. The groove portions **33** each include a guide groove **37** that allows for passage of the corresponding first fixed terminal connection pin **19b** when coupling the first fixed terminal **19** to the case **11**. The guide groove **37** has, for example, a groove width that narrows in the direction (direction **A1** indicated by arrow in FIG. 6) the first fixed terminal **19** is coupled to the case **11**. In this manner, at least part of the groove portion **33** is narrowed.

(52) The groove portion **33** includes a cutout groove **38** that provides space for the first fixed terminal connection pin **19b** when coupling of the first fixed terminal **19** to the case **11** is completed. The cutout groove **38** is continuous with, for example, the lower end of the guide groove **37**. The groove portion **33** (guide groove **37** and cutout groove **38**) extends from the upper end to the lower end in the corresponding side wall **13** of the case **11**.

(53) Measures Against Entry of Foreign Material from Groove Portion **33**

(54) As shown in FIG. 6, the cover **12** includes projecting walls **39** fitted into at least part of the groove portions **33** when the cover **12** is coupled to the case **11**. In one example, each projecting wall **39** is fitted into the corresponding groove portion **33** from the upper end to an intermediate portion in the depth direction of the groove portion **33**, for example, to the lower end of the guide groove **37**. The projecting wall **39** is formed to project from a rear surface edge of the cover **12** toward the case **11**. The two projecting walls **39** are each paired with the corresponding groove

portion **33**. The projecting walls **39** are each fitted into at least part of the corresponding groove portion **33** to limit entry of foreign material from the groove portions **33** into the housing **9**. Each projecting wall **39** of the present example is formed to fit into the guide groove **37** of the corresponding groove portion **33**. In this case, the projecting wall **39** is narrowed in conformance with the shape of the guide groove **37**.

(55) As shown in FIG. 7, the switch device **8** includes two restriction walls **40** respectively arranged at the inner sides of the groove portions **33** in the case **11**. The two restriction walls **40** of the present example are respectively paired with the two groove portions **33**. The restriction walls **40** of the present example each project from the case **11** to, for example, partially cover the inner side of the lower end of the corresponding guide groove **37** and entirely cover the corresponding cutout groove **38**. The restriction wall **40** partitions each groove portion **33** (guide groove **37**, cutout groove **38**) from the inside of the case **11** so that foreign material from the groove portion **33** does not enter the case **11**.

(56) Arrangement of Connection Pin Group

(57) As shown in FIG. 6, the movable terminal connection pin **18b** and the second fixed terminal connection pin **20b** are arranged to extend in the same direction (horizontal direction) as the first fixed terminal connection pin **19b**. In other words, the movable terminal connection pin **18b** and the second fixed terminal connection pin **20b** extend in the same projection direction as the bent first fixed terminal connection pin **19b**. This allows for surface mounting when the switch device **8** is mounted on the board **21**.

(58) In the present example, the linking portions **18c**, **20c** are also arranged to extend in the same direction (horizontal direction) as the first fixed terminal connection pin **19b**. Thus, the movable terminal connection pin **18b**, the first fixed terminal connection pin **19b**, the second fixed terminal connection pin **20b**, and the linking portions **18c**, **20c** are all arranged on the same plane.

(59) Press-Fitting Structure of First Fixed Terminal **19**

(60) As shown in FIG. 8, the switch device **8** includes a press-fitting mechanism **43** that temporarily fixes the first fixed terminal **19** to the case **11** through press-fitting. The press-fitting mechanism **43** of the present example includes a press-fitting portion **44** formed in the groove portion **33** of the case **11**. The press-fitting portion **44** of the present example is arranged in the groove portion **33** next to the guide groove **37**. In one example, the press-fitting portion **44** is arranged in the side wall **13** of the case **11** where the groove portion **33** is defined, for example, inside the case **11** at the part where the guide groove **37** is defined. The press-fitting portion **44** is a recessed portion into which a proximal end **19c** of the first fixed terminal body **19a** is pressed. The press-fitting portion **44** includes two first catches **45** that are opposed to each other in the recessed portion. Each first catch **45** has a predetermined bulging shape.

(61) The press-fitting mechanism **43** includes two second catches **46** at the proximal end **19c** of the first fixed terminal **19**. The second catches **46** are arranged on the side surfaces of the proximal end **19c** and have bulging shapes so as to be bulged alternately with the corresponding first catches **45**. The press-fitting mechanism **43** temporarily fixes the first fixed terminal **19** to the case **11** by press-fitting the proximal ends **19c** of the first fixed terminal body **19a** to the press-fitting portions **44** of the case **11**. Thus, when the first catches **45** are press-fitted to the second catches **46** of the case **11**, the first fixed terminal **19** is temporarily fixed to the case **11**.

(62) Temporary Holding of Cover **12** on Case **11**

(63) As shown in FIG. 9, the switch device **8** includes temporary holding portions **48** used to temporarily fix the cover **12**, which is coupled to the case **11** to close the opening **10**, to the case **11**. The temporary holding portions **48** of the present example each include compressible ribs **49** formed on, for example, the side surfaces of the engagement portion **15** of the case **11**.

(64) The compressible ribs **49** of the present example are arranged to set the reference position of the case **11** in the device width direction (Y-axis direction in FIG. 9) and the device longitudinal direction (X-axis direction in FIG. 9). Specifically, the compressible ribs **49** include two first

compressible ribs **49a** that set the reference position of the case **11** in the device width direction and two second compressible ribs **49b** that set the reference position of the case **11** in the device longitudinal direction. In other words, the compressible ribs **49** include the two first compressible ribs **49a** that position the case **11** at the reference position in the device width direction and the two second compressible ribs **49b** that position the case **11** at the reference position in the device longitudinal direction. The cover **12** is temporarily fixed to the case **11** by the compressible ribs **49** and then permanently fixed to the case **11** by welding (for example, laser welding).

(65) Pressing of First Fixed Terminal **19** by Cover **12**

(66) As shown in FIG. **10**, the cover **12** has a rear surface including compressible ribs **50** to press the first fixed terminal **19** from above when the cover **12** is temporarily fixed to the case **11**. In the present example, three ribs **50** are arranged on the cover **12**. The ribs **50** press the first fixed terminal **19** from above when the cover **12** is temporarily fixed to the case **11**. The ribs **50**, for example, hold the first fixed terminal **19** in place until the cover **12** is welded to the case **11**.

(67) Detail of Shape of Second Fixed Terminal **20**

(68) As shown in FIGS. **11** and **12**, the second fixed terminal **20** includes a fixed contact **53**, which is a portion that contacts the movable terminal **18**, formed by machining part of the second fixed terminal **20**. The second fixed terminal **20** contacts the movable contact **26** of the movable terminal **18** at the fixed contact **53**. The fixed contact **53** is formed by cutting and bending the base material of the second fixed terminal **20**.

(69) As shown in FIG. **11**, the fixed contact **53** is shaped to project in a bulging manner toward the movable terminal **18** from a base portion **54** of the second fixed terminal body **20a**. The fixed contact **53** is shaped to be an arch as viewed from the side (Y-axis direction in FIG. **11**). The second fixed terminal **20** includes a cutout hole **55** beneath the fixed contact **53**.

(70) Electric Structure of Switch Unit **1**

(71) As shown in FIG. **13**, the switch unit **1** includes a controller **80** that detects movement of the operating portion **4** from an output of the switch device **8**. The controller **80** is connected to the first switch device **8a** that detects movement of the operating portion **4** in the first operation direction **R1** and the second switch device **8b** that detects movement of the operating portion **4** in the second operation direction **R2**. The controller **80** of the present example is connected to the first switch device **8a** and the second switch device **8b** by first to fourth ports **P1** to **P4**.

(72) Specifically, in the first switch device **8a**, the movable terminal connection pin **18b** is connected to first port **P1**, the first fixed terminal connection pin **19b** is connected to fourth port **P4**, and the second fixed terminal connection pin **20b** is connected to second port **P2**. In the second switch device **8b**, the movable terminal connection pin **18b** is connected to third port **P3**, the first fixed terminal connection pin **19b** is connected to second port **P2**, and the second fixed terminal connection pin **20b** is connected to fourth port **P4**.

(73) As shown in FIG. **14**, when the operating portion **4** is in the neutral position, the movable terminals **18** of the first switch device **8a** and the second switch device **8b** are connected to the corresponding first fixed terminals **19**. Thus, first port **P1** is connected to fourth port **P4**, and second port **P2** is connected to third port **P3**. When the controller **80** detects this connection state, the controller **80** determines that the operating portion **4** is in the neutral position.

(74) When the operating portion **4** is moved to the first operation position in the first operation direction **R1**, the connection of the movable terminal **18** is switched from the first fixed terminal **19** to the second fixed terminal **20** only in the first switch device **8a**. Thus, movement of the operating portion **4** to the first operation position connects first port **P1**, second port **P2**, and third port **P3**. When the controller **80** detects this connection state, the controller **80** determines that the operating portion **4** has been moved to the first operation position. Thus, the movement is detected as activation of the electric parking brake switch **2**.

(75) When the operating portion **4** is moved to the second operation position in the second operation direction **R2**, the connection of the movable terminal **18** is switched from the first fixed

terminal **19** to the second fixed terminal **20** only in the second switch device **8b**. Thus, movement of the operating portion **4** to the second operation position connects first port **P1**, third port **P3**, and fourth port **P4**. When the controller **80** detects this connection state, the controller **80** determines that the operating portion **4** has been moved to the second operation position. Thus, the movement is detected as deactivation of the electric parking brake switch **2**.

Operation of Embodiment

(76) The operation of the switch device **8** in the present embodiment will now be described.

(77) As shown in FIG. **3**, in the present example, the first fixed terminal **19** is not integrated with the cover **12** and separate from the cover **12**. If, for example, the cover **12** were to be integrated with the first fixed terminal **19**, two components have to be integrated into a single component. This will lower the yield of the product since a certain number of products may not be able to meet the product requirements. In the present example, the cover **12** and the first fixed terminal **19** are separate components. This will allow the product requirements to be met more readily and improve yield.

(78) As shown in FIG. **8**, the first fixed terminal **19** is temporarily fixed to the case **11** through press-fitting. This facilitates positioning of the first fixed terminal **19** and improves the assembling working efficiency of the switch device **8**.

(79) As shown in FIG. **15**, when the first fixed terminal **19** is coupled through a method other than press-fitting, for example, a flange **60** projecting from the first fixed terminal **19** may be held between the case **11** and the cover **12**. However, in this structure, the thickness of the side wall of the switch device **8** that is denoted by **W1** will be the thicknesses of the case **11** in addition to the thickness of the first fixed terminal **19**. This will hinder device miniaturization. In contrast, the present example fits the first fixed terminal **19** into the case **11**. Thus, the thickness of the side wall of the switch device **8** is **W2** ($<W1$) that is the thickness of the case **11** (first fixed terminal **19**) alone. This is advantageous for miniaturization (downsizing) of the switch device **8**.

(80) As shown in FIG. **5**, the distal ends of the first fixed terminal connection pins **19b** are bent in advance when the first fixed terminal **19** is coupled to the case **11**. Thus, bending adjustments can be made more readily than when the first fixed terminal connection pins **19b** are bent after the first fixed terminal **19** is coupled to the case **11**. Further, the first fixed terminal connection pins **19b** are bent with higher accuracy when bent in advance. This improves the coplanarity accuracy when mounted on the board **21**.

(81) The first fixed terminal connection pins **19b**, which are bent in advance, extend sideward from the first fixed terminal body **19a**. Thus, the case **11** needs passages that allow for passage of the first fixed terminal connection pins **19b** without interfering with the first fixed terminal connection pins **19b**. In the present example, the side walls **13** of the case **11** include the groove portions **33** (specifically, guide grooves **37**) so that the first fixed terminal connection pins **19b** are allowed to pass through the side walls **13**.

(82) As shown in FIG. **6**, the cover **12** includes the projecting wall **39** to close the groove portion **33**. When the side wall **13** of the case **11** includes the groove portion **33**, foreign material may enter the case **11** from the groove portion **33**. In the present example, the cover **12** includes the projecting wall **39** to close the groove portion **33**. This limits entry of foreign material from the groove portion **33**.

(83) The extension direction of the pin group including the movable terminal connection pins **18b**, the first fixed terminal connection pins **19b**, and the second fixed terminal connection pins **20b** is parallel to the mounting surface **34** of the board **21**. Thus, the switch device **8** is mounted on the board **21** through surface mounting rather than through-hole mounting. This improves the degree of freedom for mounting and allows the switch device **8** to be reduced in size.

Advantages of Embodiment

(84) The switch device **8** in the above embodiment has the following advantages.

(85) (1) The switch device **8** includes the movable terminal **18**, the first fixed terminal **19**, and the

second fixed terminal **20**. The movable terminal **18** has the snap action mechanism **23** that bends the spring **22** when the user moves the operating portion **4**. The movable terminal **18** contacts the first fixed terminal **19** when the spring **22** is not bent.

(86) The movable terminal **18** contacts the second fixed terminal **20** when the operating portion **4** is moved by a specified amount thereby bending the spring **22**. The switch device **8** includes the case **11** that has the opening **10** and the cover **12** that closes the opening **10**. The cover **12** is separate from the first fixed terminal **19**. The cover **12** closes the opening **10** and positions the first fixed terminal **19** in the case **11**.

(87) With this structure, the cover **12** is separate from the first fixed terminal **19**. Thus, product requirements will be met more easily than when the cover **12** and the first fixed terminal **19** are integrally manufactured. This improves the yield of the components of the switch device **8**.

(88) (2) The switch device **8** includes the press-fitting mechanism **43** that temporarily fixes the first fixed terminal **19** to the case **11** through press-fitting. With this structure, when the switch device **8** is assembled, the first fixed terminal **19** can be temporarily fixed to the case **11** through press-fitting. This allows for easy positioning of the first fixed terminal **19** and facilitates assembly of the switch device **8**.

(89) (3) The switch device **8** includes the temporary holding portions **48** that temporarily fix the cover **12**, which is coupled to the case **11** to close the opening **10**, to the case **11**. With this structure, the cover **12** is temporarily held in the case **11** and then fixed to the case **11**. Thus, during fixing the cover **12** fixed to the case **11**, the cover **12** is not displaced from the case **11**. This improves the positioning accuracy when the cover **12** is fixed to the case **11**.

(90) (4) The cover **12** has the rear surface including the compressible ribs **50** to press the first fixed terminal **19** from above when the cover **12** is temporarily fixed to the case **11**. With this structure, when the cover **12** is placed on the case **11**, the ribs **50** of the cover **12** press the first fixed terminal **19** from above to position the first fixed terminal **19**. This improves the positioning accuracy of the first fixed terminal **19** relative to the case **11**.

(91) (5) The first fixed terminal **19** includes the first fixed terminal body **19a** that contacts the movable terminal **18** and the first fixed terminal connection pins **19b** arranged at the ends of the first fixed terminal body **19a** and mounted on the board **21**. The first fixed terminal connection pins **19b** are bent in advance in the direction parallel to the mounting surface **34** of the board **21** before the first fixed terminal **19** is coupled to the case **11**. With this structure, the first fixed terminal connection pins **19b** are bent in advance before the first fixed terminal **19** is coupled to the case **11**. This facilitates bending of the first fixed terminal connection pins **19b** since only the first fixed terminal **19** undergoes the bending.

Other Embodiments

(92) The present embodiment may be modified as follows. The present embodiment and the following modifications can be combined if the combined modifications remain technically consistent with each other.

(93) The first fixed terminal **19** does not need to include the connection pins at two sides. Instead, the first fixed terminal **19** may include the connection pin only at one side.

(94) The first fixed terminal connection pins **19b** do not need to be substantially L-shaped. Instead, the first fixed terminal connection pins **19b** may be shaped to, for example, extend sideward straight from the proximal end **19c**.

(95) The proximal ends **19c** of the first fixed terminal **19** do not need to be press-fitted into the case **11**. Instead, other portions of the first fixed terminal **19** such as the body portion or the connection pins may be press-fitted into the case **11**.

(96) The first fixed terminal connection pins **19b** of the first fixed terminal **19** do not need to be bent in advance. For example, the first fixed terminal connection pins **19b** may be bent after the housing **9** is assembled.

(97) The fixed contact **53** of the second fixed terminal **20** may be a rivet that is a component

separate from, for example, the base portion **54** of the second fixed terminal **20**.

(98) The movable terminal **18** may be any member having the snap action mechanism **23**.

(99) The guide groove **37** (groove portion **33**) does not need to be narrowed. Instead, the guide groove **37** may be straight or have any other shape.

(100) The groove portion **33** may include, for example, only the guide groove **37**.

(101) The groove portion **33** does not need to include the press-fitting portion **44**. That is, the press-fitting mechanism **43** may be omitted.

(102) The projecting wall **39** may entirely cover the groove portion **33**.

(103) The restriction wall **40** does not need to be arranged in the case **11**. Instead, the restriction wall **40** may be arranged on the cover **12**.

(104) The height of the restriction wall **40** may be changed.

(105) The cover **12** may be divided into segments with one of the segments structured to position the first fixed terminal **19**.

(106) The movable terminal **18**, the first fixed terminal **19**, and the second fixed terminal **20** may be formed from a single metal plate through pressing or the like.

(107) The contacts do not need to be mounted on the board **21** by planar mounting. Instead, the contacts may be mounted on the board **21** by through-hole mounting.

(108) Application of the switch device **8** is not limited to the electric parking brake. The switch device **8** may be applied to other devices or apparatuses.

(109) The switch device **8** is not limited to an onboard device and may be used for other devices, apparatuses, or systems.

(110) While the present disclosure is described with reference to examples, the present disclosure is not limited to the example or the configuration of the example. The present disclosure includes various variations and modifications within an equivalent range. In addition, various combinations and forms and other combinations and forms, which include only one element or more, shall be within the scope or a range of ideas of the present disclosure.

(111) Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. A switch device, comprising: a movable terminal including a snap action mechanism that bends a spring when a user moves an operating portion; a first fixed terminal that contacts the movable terminal when the spring is not bent; a second fixed terminal that contacts the movable terminal when the operating portion is moved by a specified amount to bend the spring; a case including an opening; a cover that closes the opening and is separate from the first fixed terminal, wherein when the opening is closed by the cover, the first fixed terminal is positioned in the case by the cover; and a press-fitting mechanism that temporarily fixes the first fixed terminal to the case when the first fixed terminal is press-fitted to the case.

2. The switch device according to claim 1, further comprising: a temporary holding portion that temporarily fixes the cover to the case, the cover being coupled to the case to close the opening.

3. The switch device according to claim 1, wherein the cover includes a rear surface including a compressible rib that presses the first fixed terminal from above when the cover is temporarily

fixed to the case.

4. The switch device according to claim 1, wherein the first fixed terminal includes: a first fixed terminal body that contacts the movable terminal; and a first fixed terminal connection pin that is arranged at an end of the first fixed terminal body and configured to be mounted on a board, and the first fixed terminal connection pin is bent in advance in a direction parallel to a mounting surface of the board before the first fixed terminal is coupled to the case.

5. The switch device according to claim 1, wherein the press-fitting mechanism temporarily fixes the first fixed terminal to the case by press-fitting proximal ends of the first fixed terminal to the case.

6. The switch device according to claim 5, wherein the first fixed terminal includes: a first fixed terminal body that contacts the movable terminal, wherein the first fixed terminal body is C-shaped, the proximal ends of the first fixed terminal are two proximal ends, which are opposed to each other, of the first fixed terminal body; and two first fixed terminal connection pins that are respectively arranged at the two proximal ends of the first fixed terminal body and configured to be mounted on a board.

7. The switch device according to claim 6, wherein each of the two first fixed terminal connection pins is bent in advance in a direction parallel to a mounting surface of the board before the first fixed terminal is coupled to the case.
